Highlights

- On NQ, dense encoders passed 0.16–0.27 EM points per Hit@5 point gained
- On the full 2.68M-passage NQ corpus, two dense encoders gave ratios of 0.15–0.23
- New NQ hits raised EM; queries that kept or lost a hit offset part of the gain
- HotpotQA ratios fell from 0.49–0.93 to 0.29–0.35 when Both@5 replaced Hit@5
- Retrieval-stage throughput on a T4 GPU differed 27-fold across six encoders